MATH3303 – Lecture 21

Multiplicative Inverses, Units, and One-Sided Inverses

Review Notes

Reference: Lee, Abstract Algebra*, Section 9.1*

Overview

This lecture develops the theory of multiplicative inverses in rings, with particular attention to subtleties that arise when a ring is non-commutative or infinite:

- **Uniqueness** of two-sided multiplicative inverses
- **Units** and the unit set $R^\times$
- **One-sided inverses** (left and right multiplicative inverses)
- **Finite rings:** one-sided invertibility implies two-sided invertibility
- **Counterexample in an infinite ring:** $\operatorname{End}(\mathbb{R}[x])$, with integration I, differentiation D, and evaluation E

Connection to Lecture 20. Recall the definition of a unit: $a \in R$ is a unit if there exists $b \in R$ with $ab = ba = 1$. In Lecture 20 we proved that for a ring with identity, the set of units $U(R) = R^\times$ forms a group under multiplication. Here we examine *when* such inverses exist, whether they are unique, and what happens if we only have one-sided inverses.

1 Multiplicative Inverses

Definition 1.1: Multiplicative inverse

Let R be a ring with identity 1, and let $a \in R$. An element $b \in R$ is called a **(two-sided) multiplicative inverse** of a if

$$ab = ba = 1.$$

Proposition 1.1: Uniqueness of the multiplicative inverse

Let R be a ring with identity. If $a \in R$ has a multiplicative inverse, then it is unique.

Proof. Suppose $b, b' \in R$ both satisfy

$$ab = ba = 1 \quad \text{and} \quad ab' = b'a = 1.$$

Then, using associativity of multiplication,

$$b = b \cdot 1 = b(ab') = (ba)b' = 1 \cdot b' = b'.$$

Hence $b = b'$. □

Remark 1.1: Notation

Because of uniqueness, when the multiplicative inverse of a exists, we write it as a^{-1}.

2 Units and the Set $R^\times$

Definition 2.1: Unit

Let R be a ring with identity. An element $a \in R$ is called a **unit** if it has a multiplicative inverse, i.e. there exists $b \in R$ such that $ab = ba = 1$. The set of all units of R is denoted

$$R^\times = U(R) = \{\, a \in R : \exists\, b \in R \text{ with } ab = ba = 1 \,\}.$$

Remark 2.1: $R^\times$ is a group

From Lecture 20 (Theorem 8.8 in Lee), $R^\times$ is a group under multiplication, called the *unit group* of R.

3 One-Sided Multiplicative Inverses

A key subtlety: in a non-commutative ring, an element may have an inverse on one side without having one on the other.

Definition 3.1: Left and right multiplicative inverses

Let R be a ring with identity, and let $a \in R$.

- An element $b \in R$ is a **left multiplicative inverse** of a if
$$ba = 1.$$
- An element $c \in R$ is a **right multiplicative inverse** of a if
$$ac = 1.$$

We say a is *left-invertible* (resp. *right-invertible*) if such a b (resp. c) exists.

Remark 3.1: Comparison

A two-sided multiplicative inverse is both a left and a right inverse. In a *commutative* ring, left and right inverses automatically coincide, so the distinction is only interesting in non-commutative rings.

4 The Finite Ring Case

In a *finite* ring with identity, the distinction between left and right inverses collapses: any one-sided inverse is automatically a two-sided inverse.

Theorem 4.1: One-sided inverses in finite rings

Let R be a finite ring with identity, and let $a \in R$. If there exists $b \in R$ such that $ba = 1$, then $ab = 1$. Similarly, $ac = 1$ implies $ca = 1$.

Proof. Let R be a finite ring with identity, and suppose $a, b \in R$ satisfy $ba = 1$. We wish to show $ab = 1$.

Step 1: Define a left-multiplication map. Define

$$\varphi_a : R \longrightarrow R, \qquad \varphi_a(x) = ax.$$

Step 2: φ_a is injective. Suppose $\varphi_a(x) = \varphi_a(y)$ for some $x, y \in R$. Then $ax = ay$, and multiplying on the left by b:

$$b(ax) = b(ay) \implies (ba)x = (ba)y \implies 1 \cdot x = 1 \cdot y \implies x = y.$$

So φ_a is injective.

Step 3: Injectivity implies surjectivity on a finite set. Since R is finite and $\varphi_a : R \to R$ is an injective map from a finite set to itself, it is also *surjective* (and hence bijective).

Step 4: Find a right inverse. By surjectivity, there exists $c \in R$ with $\varphi_a(c) = 1$, i.e. $ac = 1$.

Step 5: Show $b = c$. We have $ba = 1$ and $ac = 1$. Then

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c.$$

So $b = c$, and therefore $ab = ac = 1$. Combined with $ba = 1$, this shows b is a two-sided multiplicative inverse of a. □

Remark 4.1: What goes wrong if R is infinite?

The proof depends crucially on Step 3: an injective map from a finite set to itself is automatically surjective. For an infinite set this is false (e.g. the shift map $n \mapsto n + 1$ on $\mathbb{Z}$ is injective but not surjective). So in an infinite ring, φ_a may be injective without being surjective, and a left inverse need not be a right inverse. The next section gives an explicit example.

5 An Infinite Ring Where Left Inverses Are Not Unique

We now exhibit an infinite ring with identity in which a single element has *two distinct left multiplicative inverses*. This shows that the uniqueness result (Proposition 1.1) really does require a two-sided inverse.

Example 5.1: Endomorphisms of $\mathbb{R}[x]$

Let $R = \mathrm{End}(\mathbb{R}[x])$ denote the ring of $\mathbb{R}$-linear maps $T : \mathbb{R}[x] \to \mathbb{R}[x]$, with

- **Addition:** $(S + T)(f) = S(f) + T(f)$
- **Multiplication (composition):** $(S \cdot T)(f) = S(T(f))$
- **Identity:** the identity map $\mathrm{id} : \mathbb{R}[x] \to \mathbb{R}[x]$, $\mathrm{id}(f) = f$

Define three specific linear maps on $\mathbb{R}[x]$:

(1) Integration $I : \mathbb{R}[x] \to \mathbb{R}[x]$:

$$I(f)(x) = \int_0^x f(t)\,dt.$$

On monomials: $I(x^n) = \dfrac{x^{n+1}}{n+1}$ for $n \geq 0$. In particular $I(1) = x$, $I(x) = x^2/2$, etc.

(2) Differentiation $D : \mathbb{R}[x] \to \mathbb{R}[x]$:

$$D(f)(x) = f'(x).$$

On monomials: $D(x^n) = nx^{n-1}$.

(3) Evaluation at 0 $E : \mathbb{R}[x] \to \mathbb{R}[x]$:

$$E(f)(x) = f(0) \quad \text{(the constant polynomial equal to the constant term of } f\text{)}.$$

On monomials: $E(1) = 1$ and $E(x^n) = 0$ for $n \geq 1$.

Two left inverses of I**.** We compute $D \circ I$ and $(D + E) \circ I$:

First inverse. By the Fundamental Theorem of Calculus,

$$(D \circ I)(f)(x) = D\left(\int_0^x f(t)\,dt\right) = f(x).$$

So $D \circ I = \mathrm{id}$. That is, D is a *left multiplicative inverse* of I in $\mathrm{End}(\mathbb{R}[x])$.

Second inverse. Next, observe that

$$(E \circ I)(f)(x) = E\left(\int_0^x f(t)\,dt\right) = \int_0^x f(t)\,dt\bigg|_{x=0} = 0.$$

So $E \circ I = 0$ (the zero map). Therefore

$$(D + E) \circ I = D \circ I + E \circ I = \mathrm{id} + 0 = \mathrm{id}.$$

So $D + E$ is *also* a left multiplicative inverse of I.

D **and** $D + E$ **are different.** Evaluate both on the constant polynomial $f(x) = 1$:

$$\begin{aligned} D(1) &= 0, \\ (D + E)(1) &= D(1) + E(1) = 0 + 1 = 1. \end{aligned}$$

Thus $D \neq D + E$, and I has two distinct left multiplicative inverses in $\mathrm{End}(\mathbb{R}[x])$.

Remark 5.1: Why doesn't this contradict Proposition 1.1?

Proposition 1.1 concerns *two-sided* multiplicative inverses, which are unique whenever they exist. In the example above, neither D nor $D + E$ is a two-sided inverse of I: indeed, I is not surjective in $\mathrm{End}(\mathbb{R}[x])$ (its image consists of polynomials with zero constant term), so I has no right inverse and hence no two-sided inverse. The example shows that the hypothesis "two-sided" is essential.

Remark 5.2: Why doesn't this contradict Theorem 4.1?

That theorem requires R to be *finite*. The ring $\mathrm{End}(\mathbb{R}[x])$ is infinite-dimensional over $\mathbb{R}$, so the argument via injective $\Rightarrow$ surjective breaks down.

6 Summary

Setting	Left $\Rightarrow$ Two-sided?	Inverse unique?
General ring with identity	No	Yes if two-sided exists
Commutative ring with identity	Yes (left = right)	Yes
Finite ring with identity	Yes	Yes
$\mathrm{End}(\mathbb{R}[x])$ (infinite, non-commutative)	No (counterexample: I)	No (for one-sided)

Key takeaways.

1. Two-sided inverses are always unique (when they exist).
2. In finite rings, one-sided inverses are automatically two-sided.
3. In infinite, non-commutative rings, one-sided inverses can exist and may not be unique; the integration operator I on $\mathbb{R}[x]$ is the canonical example.
4. The proof technique "injective on a finite set is surjective" is the engine behind the finite-ring result.

7 Practice Problems

Exercise 7.1: Inverses in $\mathbb{Z}_n$

Determine the unit group $\mathbb{Z}_{12}^{\times}$. For each unit, find its multiplicative inverse.

Exercise 7.2: Right inverse in a finite ring

Let R be a finite ring with identity, and suppose $a, c \in R$ with $ac = 1$. Show that $ca = 1$. (Mirror the proof of Theorem 4.1.)

Exercise 7.3: Polynomial endomorphisms

In $\mathrm{End}(\mathbb{R}[x])$, find a third left inverse of I different from D and $D + E$.
Hint. Try $D + cE$ for any $c \in \mathbb{R}$, or compose E with other constant-producing maps.

Exercise 7.4: Direct sum of units

Let R and S be rings with identity. Show that $(r, s) \in R \oplus S$ is a unit if and only if $r \in R^{\times}$ and $s \in S^{\times}$. Conclude that $(R \oplus S)^{\times} = R^{\times} \times S^{\times}$. (*Lee, Exercise 8.29*)

Exercise 7.5: Idempotents in integral domains

Show that every integral domain contains exactly two elements a satisfying $a^2 = a$. (*Lee, Exercise 8.31*)

Exercise 7.6: Matrix rings

In $\mathrm{M_n}(\mathbb{R})$, give an example of a matrix that has a left inverse but no right inverse (or explain why no such example exists).

Exercise 7.7: Shift operator

Let $V = \mathbb{R}[x]$, and define the linear map $S : V \to V$ by $S(f)(x) = xf(x)$ (multiplication by x). Show that S has a left inverse but no right inverse, and find two distinct left inverses.

Checklist of skills

- ☐ State the definition of a multiplicative inverse and prove uniqueness.
- ☐ Distinguish left, right, and two-sided multiplicative inverses.
- ☐ Prove that in a finite ring with identity, $ba = 1$ implies $ab = 1$.
- ☐ Construct the integration / differentiation / evaluation maps on $\mathbb{R}[x]$.
- ☐ Verify $D \circ I = \mathrm{id}$ and $E \circ I = 0$ from the definitions.
- ☐ Explain why D and $D + E$ are distinct left inverses of I.
- ☐ Identify which hypotheses in the finite-ring theorem are essential.